

Topic 5: Alkenes

Related topics in Units 2, 4 and 5 will assume knowledge of this material.

Students will be assessed on their ability to:

5.1	know the general formula of alkenes and understand that alkenes and cycloalkenes are hydrocarbons which are unsaturated (have a carbon-carbon double bond which consists of a σ bond and a π bond)
5.2	be able to explain geometric isomerism in terms of restricted rotation around a C=C double bond and the nature of the substituents on the carbon atoms
5.3	understand the <i>E-Z</i> naming system for geometric isomers and why it is necessary to use this when the <i>cis-</i> and <i>trans-</i> naming system breaks down
5.4	be able to describe the reactions of alkenes, limited to: i the addition of hydrogen, using a nickel catalyst, to form an alkane ii the addition of halogens to produce a di-substituted halogenoalkane iii the addition of hydrogen halides to produce mono-substituted halogenoalkanes iv the addition of steam, in the presence of an acid catalyst, to produce alcohols v oxidation of the double bond by acidified potassium manganate(VII) to produce a diol
5.5	know the qualitative test for a C=C double bond using bromine or bromine water
5.6	be able to describe the mechanism (including diagrams), giving evidence where possible, of: i the electrophilic addition of bromine and hydrogen bromide to ethene ii the electrophilic addition of hydrogen bromide to propene <i>Use of the curly arrow notation is expected – the curly arrows should start from either a bond or from a lone pair of electrons.</i> <i>Knowledge of the relative stability of primary, secondary and tertiary carbocation intermediates is expected.</i>
5.7	be able to describe the addition polymerisation of alkenes and draw the repeat unit given the monomer, and vice versa
5.8	understand how chemists limit the problems caused by polymer disposal by: i developing biodegradable polymers ii removing toxic waste gases produced by the incineration of polymers
	Further suggested practicals: i investigating the difference in reactivity of alkanes and alkenes, including combustion, reaction with bromine water, reaction with acidified potassium manganate(VII) ii preparation of cyclohexene from cyclohexanol iii preparation of limonene from orange peel by steam distillation iv preparation of Perspex® from methyl 2-methylpropenoate